From Three-Layer Unlearning Impact to Forget-Set Audit: Predicting Impact Profiles from Forget-Set Geometry Alone

Anonymous ACL submission

Abstract

Large language model (LLM) unlearning must remove target knowledge represented by a forget set $\mathcal{D}_f$ while limiting collateral damage to useful knowledge. Existing evaluations usually collapse this requirement into aggregate forget/retain scores or compare algorithms on a fixed forget split, leaving unclear how unlearning impact spreads from the target set to neighbouring and unrelated text, and whether damaging forget sets can be identified before running unlearning. We construct 100 WikiText-103 triplets, unlearn Llama-3.1-8B-Instruct with GradAscent for each forget set, and evaluate all checkpoints against all triplets in a 100×100 cross-PPL matrix, yielding 1.5M per-sample perplexity ratios. We represent each run by a *three-layer impact profile*: intended degradation on $\mathcal{D}_f$ (L_1), neighbour impact on held-out same-triplet samples (L_2), and bystander impact on off-diagonal test samples (L_3); we then predict this profile from a 12-feature intrinsic geometry descriptor of $\mathcal{D}_f$ using a leave-one-out Random Forest auditor. Impact decays across layers (L_1=1.96×, L_2=1.32×, L_3=1.19×), but spillover does not vanish: 73.8% of off-diagonal bystander samples have elevated PPL, and L_1 varies by 1.59× across forget sets under the same algorithm and hyperparameters. The geometry-only auditor gives rank-significant predictions for all layers (Spearman ρ=0.56/0.87/0.59 for $L_1/L_2/L_3$), and an independent NPO replication remains rank-significant (ρ=0.66/0.82/0.51, all CIs above zero). Unlearning side effects are partly a property of the forget set itself, and forget-set geometry can provide a cheap pre-screen for risky candidates before full unlearning evaluation.

1 Introduction

After deployment, large language models are increasingly asked to remove specific knowledge. Copyright disputes, privacy regulation, user deletion requests, and the discovery of toxic, private, or copyrighted content in pretraining corpora all create cases where a model should stop reproducing or relying on a target forget set $\mathcal{D}_f$ (Maini et al., 2024; Shi et al., 2024; Dorna et al., 2025). Since retraining from scratch is usually impractical, *LLM unlearning* aims to suppress this target knowledge through post-training updates while preserving the rest of the model. This preservation requirement is critical: in deployed systems, a successful deletion is not sufficient if the same update silently corrupts nearby facts, paraphrases, or unrelated capabilities.

Existing unlearning evaluations do not yet expose this side effect with enough resolution. Most studies report aggregate utility or a small set of post-hoc probes, which can miss the *geometry* of unlearning damage: how impact changes as evaluation text moves from the forget set itself, to same-topic neighbours, to off-target bystanders. Recent knowledge-hole probing work shows that benchmark-level utility can remain high even when adjacent or latent knowledge is corrupted (Ko et al., 2025). At the same time, most benchmarks fix the forget split and compare algorithms, treating the choice of $\mathcal{D}_f$ as given. This leaves a practical question unanswered: among many candidate forget sets, can we identify which ones are likely to cause broader damage before running expensive unlearning?

Research questions.
RQ1. Does collateral damage vary systematically with the choice of forget set?
RQ2. If yes, can it be predicted from forget-set geometry alone, before running unlearning?

We address this question with a controlled cross-evaluation framework for three-layer unlearning

impact. We cluster WikiText-103 passages into semantic groups, sample 100 candidate triplets, and split each triplet into disjoint train, validation, and test texts. For each triplet, the train split is the forget set $\mathcal{D}_f$, while validation and test texts provide held-out same-topic neighbours. After unlearning Llama-3.1-8B-Instruct with a fixed GradAscent configuration, we evaluate every unlearned checkpoint on every triplet, yielding a cross-PPL matrix with 1.5M per-sample perplexity ratios. This matrix separates **intended degradation** on the forget set (L_1), **locality damage** on same-topic held-out text (L_2), and **spillover damage** on off-diagonal bystanders (L_3). The core insight is that unlearning impact is structured by semantic distance, but the chosen forget set also has an intrinsic geometry that helps predict how severe that impact will be.

Our measurements show both distance decay and persistent collateral damage. Across the cross-PPL matrix, impact decreases strictly across the three layers ($L_1 > L_2 > L_3$), yet it does not vanish: L_3 remains above the base model on 73.8% of off-diagonal bystander samples. Damage magnitude also varies substantially with the chosen forget set, even under the same algorithm and hyperparameters; L_1 varies by $1.59\times$ across the 100 candidates. We then formulate *forget-set audit*: predicting the three-layer impact profile from forget-set geometry alone. Using a 12-dimensional intrinsic geometry descriptor computed from the 50 forget-set embeddings, a Random Forest regressor obtains rank-significant predictions under leave-one-out cross-validation, with Spearman correlations of 0.56, 0.87, and 0.59 for L_1, L_2, and L_3; all 95% bootstrap confidence intervals exclude zero. The auditor runs in seconds and serves as a pre-screen for forget-set selection before full unlearning.

Our contributions are: (1) We introduce a measurement framework for three-layer unlearning impact, separating intended forget-set degradation, same-topic locality damage, and off-diagonal spillover damage within a unified cross-PPL matrix. (2) We provide quantitative evidence that unlearning impact decays with semantic distance, yet collateral damage remains non-zero even far from the forget set: $L_3 > 1$ on 73.8% of off-diagonal bystander samples. (3) We formulate *forget-set audit*, which predicts the three-layer impact profile from forget-set geometry alone. A 12-feature Random Forest auditor achieves rank-significant predictions across all three layers, including $\rho = 0.87$ and $R^2 = 0.74$ for locality damage, enabling cheap screening before full unlearning.

Sec. 2 formalises the three-layer impact and audit problems. Sec. 3 describes the triplet schema, unlearning procedure, and cross-PPL matrix. Sec. 4 reports the three-layer impact findings. Sec. 5 presents the audit features, forget-set audit, and predictor. Sec. 6 discusses mechanisms, failure modes, and practical considerations. Sec. 7 concludes, and Sec. 8–9 cover related work and limitations.

2 Problem Formulation

Experimental objects. Let M_{θ_0} denote the original language model before unlearning, and let $\mathcal{U}$ be a fixed post-training unlearning algorithm. We consider a collection of candidate semantic groups $\mathcal{G} = \{G_i\}_{i=1}^{N}$, where each group contains passages about a coherent topic. Each group is split into three disjoint subsets: G_i^{tr}, G_i^{val}, and G_i^{te}. For candidate i, the forget set and resulting unlearned checkpoint are

$$\mathcal{D}_f^{(i)} = G_i^{\text{tr}}, \tag{1}$$

$$M_{\theta_i} = \mathcal{U}(M_{\theta_0}, \mathcal{D}_f^{(i)}). \tag{2}$$

The experiment fixes M_{θ_0} and $\mathcal{U}$, varies only $\mathcal{D}_f^{(i)}$, and compares the resulting checkpoints across the same semantic groups.

Measurement metric. For an evaluation passage x, we measure unlearning impact by the base-normalised perplexity ratio

$$r_i(x) = \frac{\text{PPL}(M_{\theta_i}, x)}{\text{PPL}(M_{\theta_0}, x)}, \tag{3}$$

$$R_i(S) = \exp\left(\frac{1}{|S|}\sum_{x\in S}\log r_i(x)\right). \tag{4}$$

A value $r_i(x) > 1$ means that unlearning $\mathcal{D}_f^{(i)}$ makes the model assign lower likelihood to x than the original model did. For a set of passages S, $R_i(S)$ is the geometric mean ratio. Normalising by M_{θ_0} makes measurements comparable across passages with different base difficulty.

Three-layer impact profile. For each candidate forget set $\mathcal{D}_f^{(i)}$, we separate impact into three semantic layers. Let $\mathcal{N}^{(i)} = G_i^{\text{val}} \cup G_i^{\text{te}}$ be the held-out same-triplet neighbour set, and let $\mathcal{B}^{(i)} =$

$\bigcup_{j \neq i} G_j^{\text{te}}$ be the off-diagonal bystander set. The three-layer impact profile is

$$L_1^{(i)} = R_i(G_i^{\text{tr}}) \qquad \text{(intended degradation)}, \tag{5}$$

$$L_2^{(i)} = R_i(\mathcal{N}^{(i)}) \qquad \text{(locality damage)}, \tag{6}$$

$$L_3^{(i)} = R_i(\mathcal{B}^{(i)}) \qquad \text{(off-diagonal spillover)}, \tag{7}$$

$$\mathbf{y}^{(i)} = \left(L_1^{(i)}, L_2^{(i)}, L_3^{(i)}\right). \tag{8}$$

High L_1 is the intended forgetting signal; high L_2 or L_3 indicates collateral damage.

Forget-set audit. Measuring $\mathbf{y}^{(i)}$ requires running $\mathcal{U}$ and evaluating the resulting checkpoint. We formulate *forget-set audit* as a pre-unlearning prediction task: given only the candidate forget set $\mathcal{D}_f^{(i)}$, predict its future three-layer impact profile. Let $\phi(\mathcal{D}_f^{(i)}) \in \mathbb{R}^d$ be an intrinsic descriptor computed from the forget-set passages. An auditor $A_\psi : \mathbb{R}^d \to \mathbb{R}^3$ is trained with the objective

$$\hat{\psi} = \arg\min_{\psi} \sum_{i=1}^{N} \left\| A_\psi\left(\phi(\mathcal{D}_f^{(i)})\right) - \mathbf{y}^{(i)} \right\|_2^2. \tag{9}$$

At prediction time, the auditor has no access to M_{θ_i} or to post-unlearning evaluation results. Its practical goal is to estimate or rank which candidate forget sets are likely to induce larger intended degradation, same-topic locality damage, or off-diagonal spillover damage before paying for full unlearning and cross-evaluation.

3 Experimental Setup

3.1 Triplet Schema

We start from WikiText-103 (Merity et al., 2017), filter to 16,038 usable passages, and embed them with `all-MiniLM-L6-v2` (Wang et al., 2021) into 384-dimensional vectors. HDBSCAN (Campello et al., 2013) with `min_cluster_size=200`, `min_samples=5`, Euclidean distance, and Excess-of-Mass (EOM) cluster selection produces 10 non-noise semantic clusters and a noise pool of 9,198 passages (57.4%). We discard the noise pool and use only the clustered passages when constructing forget-set candidates. Fig. 1 summarises the construction.

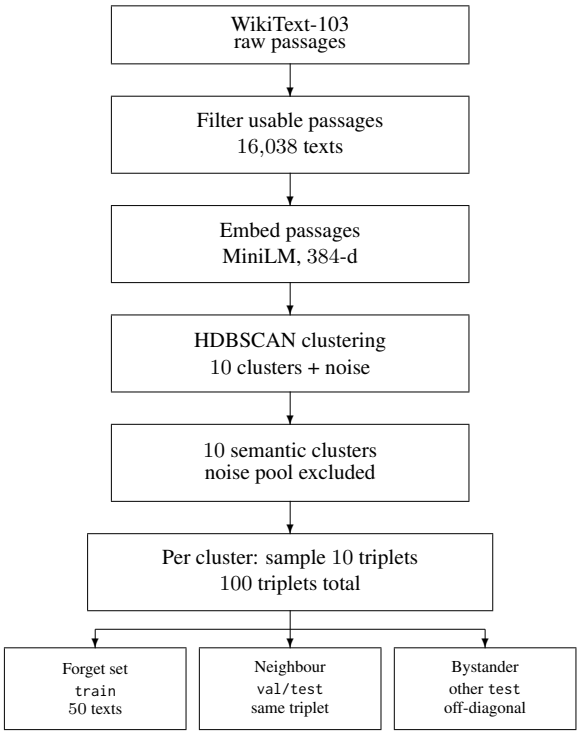

Figure 1: Triplet schema. WikiText-103 passages are filtered, embedded, and clustered; each non-noise semantic cluster yields 10 triplets. The train split is the forget set, validation/test splits are same-topic neighbours, and test splits from triplets in other clusters serve as cross-topic bystanders in off-diagonal evaluation.

For each non-noise cluster we independently sample 10 *triplets*, giving 100 triplets in total. Each triplet contains three same-cluster but mutually disjoint splits of 50 texts:

Layer	Downstream role
L_1: forget set $\mathcal{D}_f$	unlearning target
L_2: neighbours	neighbour knowledge
L_3: bystanders	unintended knowledge corruption

The within-triplet disjointness gives a clean separation between the unlearning target, same-topic neighbour knowledge, and off-diagonal corruption probes. Triplets are sampled independently, so two triplets drawn from the same cluster may reuse passages; triplets from different clusters are disjoint by construction because HDBSCAN assigns each passage to a single cluster. In the cross-PPL matrix, the matched validation and test splits instantiate L_2, while off-diagonal test splits instantiate L_3.

3.2 Unlearning Procedure

We unlearn Llama-3.1-8B-Instruct (Dubey et al., 2024) with the GradAscent objective (Jang et al., 2023): for each $\mathcal{D}_f$ we maximise the negative log-likelihood on $\mathcal{D}_f$ while keeping a small retain-loss term on the same-cluster validation split. Hyperparameters follow defaults from OpenUnlearning

(Dorna et al., 2025): learning rate 1×10^{-5} with linear decay, paged AdamW (32-bit), per-device batch size 8 with gradient accumulation 8 (effective batch 64), 5 epochs (resulting in 5 optimiser steps because $|\mathcal{D}_f|$=50 $<$ 64), warmup of 1 step, weight decay 0.01, BF16. All runs use a single H100 80 GB. Each unlearn run takes $\approx$ 75 s wall-clock; the full n=100 sweep finishes in 1 h 54 min and produces $\approx$ 1.5 TB of checkpoints ($\approx$ 15 GB each). Train loss converges in $[-0.84, -0.62]$ (negative is expected for GradAscent).

3.3 Cross-PPL Evaluation Protocol

To attribute observed spread cleanly to $\mathcal{D}_f^{(i)}$, every unlearned checkpoint M_{θ_i} is evaluated on the *same* grid of 100 triplets $\times$ 3 splits $\times$ 50 texts ($100 \times 100 \times 3 \times 50 = 1.5$M per-sample ratios), rather than each checkpoint choosing its own bystander pool. This unified probe set ensures that differences in $(L_1^{(i)}, L_2^{(i)}, L_3^{(i)})$ across i cannot be attributed to differences in the evaluation texts. Per-sample PPLs are normalised by the matching base-model PPL (Sec. 2), so absolute-difficulty variation across passages cancels and ratios are comparable across the 1.5M-row pool. We aggregate ratios per layer by geometric mean and by the fraction of samples with r>1.1. Base-model PPLs are cached once on the 100×3 probe set; the full unlearned sweep takes $\approx$ 6 h 11 min on one H100 ($\approx$ 2.2 s per checkpoint–triplet pair), and is exported as a long-format table for downstream auditing.

4 Three-Layer Unlearning Impact

This section reports the main empirical shape of unlearning impact. Following Sec. 2, we separate the effect of a single unlearning run into three layers: L_1 is the intended degradation on the forget set, L_2 is the impact on held-out same-topic neighbours, and L_3 is the impact on cross-topic bystanders. Table 1 reports geometric means of the per-sample PPL ratio r across the 1.5M evaluations, and Fig. 2 visualises the same pattern.

We also examine whether the profile is fixed by the algorithm, or whether it depends on which forget set is chosen. Under identical optimisation, switching $\mathcal{D}_f$ changes the measured profile substantially: Table 2 contrasts the mildest and worst forget set among 100, and Fig. 3 shows the full distribution.

Together, these results support three observations.

	L_1	L_2	L_3
Samples	5,000	10,000	495,000
Geo-mean r	1.96×	1.32×	1.19×
% samples with r>1.1	100.0	90.8	73.8
% samples with r>2.0	42.0	2.4	0.8

Table 1: Three-layer unlearning impact headline (n=100 forget sets, 1.5M per-sample ratios). Impact decays strictly (L_1>L_2>L_3), but bystander impact remains measurable: L_3>1 on 73.8% of cross-topic samples.

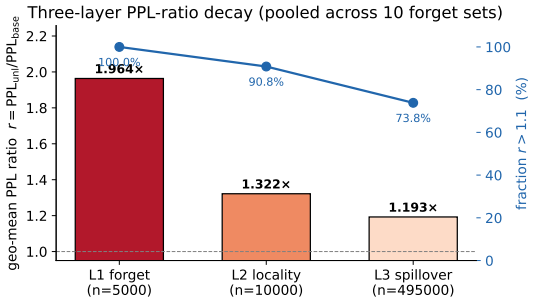

Figure 2: **Three-layer impact decay.** Left: geo-mean PPL ratio r falls from L_1=1.96× to L_3=1.19×. Right: fraction of samples with r>1.1 (100%/90.8%/73.8%). L_3 bystander impact is non-zero in both senses.

(C1) The three layers are strictly decreasing. The layer ordering is not an arbitrary binning choice: the measured impact decreases monotonically from the forget set to neighbours to bystanders. Table 1 gives L_1=1.96 × >L_2=1.32 × >L_3=1.19×, and Fig. 2 shows the same decay in both geo-mean PPL ratio and the fraction of samples with r>1.1. The ordering also holds within each cluster (Appendix B), so it is not an artefact of global aggregation.

(C2) Bystander impact is not zero. The third layer is weaker than the first two, but cross-topic bystanders are still measurably hit. L_3 has geo-mean 1.19×, and 73.8% of cross-topic samples exceed r>1.1. These samples are not in the forget set and are not same-topic neighbours of the unlearned triplet, so this is genuine spillover rather than intended forgetting.

(C3) The profile varies with the forget set. The same algorithm and hyperparameters do not produce a single fixed impact profile. In Table 2, changing only $\mathcal{D}_f$ moves L_1 from 1.54× to 2.44× and L_2 from 1.15× to 1.73×; Fig. 3 shows that this is a distributional pattern rather than a single outlier. This variation makes the forget set itself

	L_1	L_2	L_3
Mildest triplet (`029`)	1.54	1.15	1.12
Worst triplet (`073`, "storm")	**2.44**	**1.73**	**1.26**
Spread (max/min, n=100)	**1.59**×	1.50×	1.13×

Table 2: Forget-set-induced variation. L_1 varies by 1.59× between the mildest and worst $\mathcal{D}_f$ under fixed algorithm/hyperparameters.

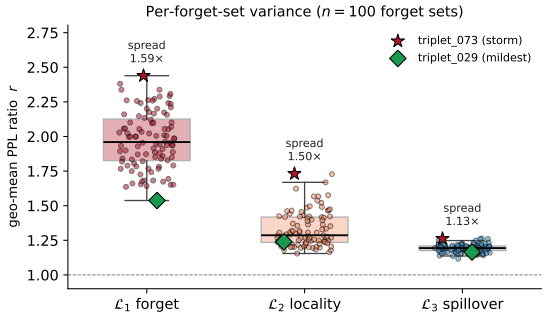

Figure 3: **Per-forget-set variance** (n=100)**.** Box + jitter of geo-mean r for each layer; one dot per forget set. Spread $\max / \min$ reaches 1.59× on L_1, 1.50× on L_2, 1.13× on L_3. `triplet_073` ("storm") is the worst on L_1 and L_2 but not on L_3 (where `triplet_072` is the maximum, 1.26×); `triplet_029` is the L_1 minimum. Distribution shape is markedly tighter as the layer recedes from $\mathcal{D}_f$.

worth auditing before paying for full unlearning and cross-evaluation.

5 Forget-Set Audit

Because the three-layer profile varies with $\mathcal{D}_f$, forget-set choice becomes a tunable design decision. The problem is that *evaluating* that decision currently requires running unlearning. We now ask whether $\mathcal{D}_f$'s intrinsic geometry is a sufficient signal to predict the three-layer impact profile without ever fine-tuning.

5.1 Audit Features

For each forget set $\mathcal{D}_f$ of 50 texts, we compute a 12-dim intrinsic geometric descriptor from the same MiniLM embeddings used for clustering, with no access to model weights, gradients, or any unlearned checkpoint. The features fall into four groups:

Group	Feature
Spread (3)	`emb_variance_mean`
	`emb_variance_max`
	`pairwise_eucl_mean`
Similarity (3)	`pairwise_sim_mean`
	`pairwise_sim_std`
	`pairwise_sim_q90`
Location/scale (3)	`centroid_norm`
	`emb_norm_mean`
	`emb_norm_std`
Shape (3)	`effective_rank`
	`isotropy`
	`spread_over_centroid`

All 12 features are scalars derived from the 50 × 384 forget-set embedding matrix; computing them takes milliseconds per $\mathcal{D}_f$.

5.2 Audit Problem

Input. For a candidate forget set $\mathcal{D}_f$, the auditor observes only the embedding-derived feature vector $\phi(\mathcal{D}_f) \in \mathbb{R}^{12}$ from Sec. 5.1. It does not use an unlearned checkpoint, evaluation texts, retain texts, gradients, or model weights.

Target. The output is a three-dimensional prediction for the future impact profile: geo-mean $\log r$ on L_1, L_2, and L_3. Equivalently, exponentiating each coordinate gives the predicted layer-level PPL ratio. We evaluate the auditor under two complementary task views: (i) *regression* (numeric fit) — how close the predicted geo-mean $\log r$ is to the true value, scored by held-out R^2 and MAE against the LOO-mean baseline; and (ii) *ranking* — whether the auditor correctly orders the 100 candidate forget sets by their future damage on each layer, scored by Spearman ρ with bootstrap CIs. Regression is the underlying learning objective; ranking is the practical use case (pre-screen the riskiest $\mathcal{D}_f$).

Training protocol. We train one Random Forest regressor per layer (n_{est}=200, minimum leaf size 2, unbounded depth, seed 0). Evaluation is leave-one-out over the n=100 forget sets: for each held-out candidate, the auditor is fit on the other 99 candidates and then predicts the held-out impact profile. The baseline predicts the mean target value from the 99 training candidates. Ridge, Lasso, and Gradient Boosting variants use the same features and LOO protocol and are reported in Sec. 5.4.

5.3 Audit Performance

Table 3 reports held-out R^2, MAE, and Spearman ρ, plus 95% percentile bootstrap CIs for ρ

	L_1	L_2	L_3
Audit R^2	$\mathbf{+0.301}$	$\mathbf{+0.735}$	$\mathbf{+0.305}$
Mean R^2	-0.020	-0.020	-0.020
Audit MAE	**0.133**	**0.046**	**0.019**
Mean MAE	0.169	0.105	0.022
Spearman ρ	$+0.560$	$+0.869$	$+0.592$
ρ 95% CI	[.40, .69]	[.81, .91]	[.44, .71]

Table 3: Forget-set audit performance. LOO over n=100 forget sets; Random Forest (n_{est}=200, min_leaf=2) on 12-dim forget-set geometry. Mean rows are LOO mean baselines. All three ρ CIs exclude zero; L_2 is the strongest layer, and L_3 CI is now strictly above zero.

(n_{boot}=10,000, seed 0, bootstrapping over the 10 LOO (true, predicted) cluster pairs).

L_2 **is the strongest layer.** With R^2=0.74 and ρ=0.87, the auditor recovers 74% of locality variance from forget-set geometry alone — consistent with the intuition that locality damage tracks how concentrated and self-similar $\mathcal{D}_f$ is in embedding space.

L_3 **CI now excludes zero.** On a coarser problem (n=10 in early experiments) the L3 audit had $R^2 = -0.46$; at n=100 the Random Forest auditor reaches $R^2 = +0.30$ with ρ=0.59 (95% CI [+0.44, +0.71]). Statistical power and a non-linear model together close the previous gap; even the linear Ridge baseline (Sec. 5.4) reaches ρ=0.49 on the same n=100 slice.

Audit cost. Computing the 12-dim feature for one $\mathcal{D}_f$ takes milliseconds; fitting the Random Forest on 99 training triplets is well under a second; predicting on a new $\mathcal{D}_f$ is essentially instant. By contrast, evaluating one candidate $\mathcal{D}_f$ via the standard pipeline costs one unlearn run ($\approx$ 75 s on H100) plus one column of cross-PPL ($\approx$ 220 s for 100 eval pairs). At n=100 candidates the full grid is $\approx$ 7 GPU-hours; the auditor runs in seconds on a CPU.

5.4 Predictor Ablation

We compare the headline Random Forest auditor against three alternative regressors on the same 12-dim feature matrix and same LOO protocol (Table 4): Ridge (α=1, the simplest linear baseline), Lasso (α=0.01), and Gradient Boosting (n_{est}=200, depth=3, lr=0.05).

The non-linear models (RF, GB) tie on the strongest layer (L_2), but RF edges out on L_1 and L_3 where forget-set geometry interactions appear to matter more. The linear Ridge baseline (ρ=0.49 on L_3 with CI [+0.33, +0.63], also strictly above zero) already gives a deployable pre-screen, indicating that a substantial fraction of the audit signal is linear in the geometric descriptor.

Predictor	L_1	L_2	L_3
R^2 (LOO held-out)			
Random Forest (headline)	$\mathbf{+0.301}$	$\mathbf{+0.735}$	$\mathbf{+0.305}$
Gradient Boosting	$+0.209$	$+0.743$	$+0.200$
Ridge (α=1)	$+0.246$	$+0.705$	$+0.215$
Lasso (α=0.01)	$+0.224$	$+0.644$	$+0.088$
Spearman ρ			
Random Forest (headline)	$\mathbf{+0.560}$	$\mathbf{+0.869}$	$\mathbf{+0.592}$
Gradient Boosting	$+0.524$	$+0.879$	$+0.534$
Ridge (α=1)	$+0.531$	$+0.840$	$+0.491$
Lasso (α=0.01)	$+0.494$	$+0.849$	$+0.380$

Table 4: Predictor ablation on the same 12-dim feature matrix and LOO protocol (n=100). RF and GB are the strongest non-linear models and tie on L_2; the linear Ridge baseline already lifts all three layers above the LOO-mean baseline (R^2=-0.02). Lasso under-fits on L_3 (sparse solution discards most features). Choosing the non-linear ceiling matters most for L_3, where RF gains $+0.09$ in R^2 over Ridge.

Metric	Alg.	L_1	L_2	L_3
geo r	GradAscent	1.96×	1.32×	1.19×
geo r	NPO	1.68×	1.15×	1.11×
ρ	GradAscent	$+0.560$	$\mathbf{+0.869}$	$+0.592$
ρ	NPO	$\mathbf{+0.657}$	$+0.823$	$+0.506$
CI	GradAscent	[.40, .69]	[.81, .91]	[.44, .71]
CI	NPO	[.52, .76]	[.72, .89]	[.34, .65]
top-10	GradAscent	2/10	6/10	3/10
top-10	NPO	2/10	**8/10**	**5/10**

Table 5: NPO replication and cross-unlearner audit. Same 12-dim geometric features, RF protocol, and LOO over n=100 forget sets; only the unlearning algorithm changes. NPO produces ∼10–14% milder impact than GradAscent, but all six ρ CIs remain strictly above zero.

5.5 Unlearner Generality (NPO)

To probe whether the audit framework is bound to GradAscent, we replicate the full pipeline with NPO (Zhang et al., 2024): a different unlearning algorithm with the same TOFU-aligned hyperparameters on the same n=100 forget sets, yielding a parallel 100 × 100 cross-PPL grid and the same 12-dim geometric descriptor as input. The audit predictor (Random Forest, identical hyperparameters and LOO protocol) is trained and evaluated independently on NPO labels we do not transfer predictions across unlearners.

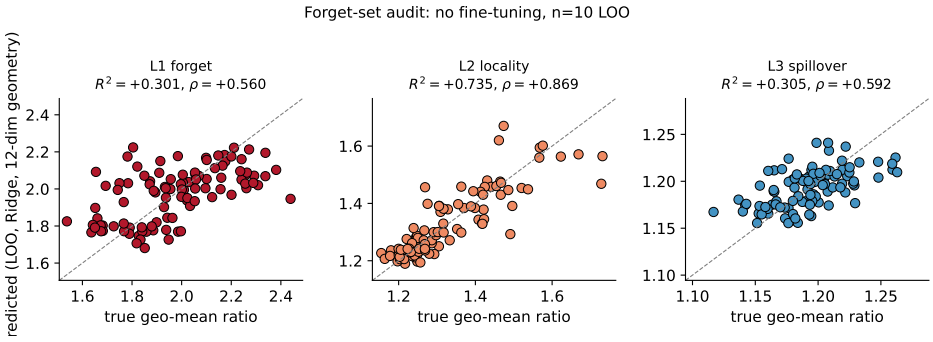

Figure 4: **Audit scatter**, predicted vs. true geo-mean $\log r$. Each point is one held-out forget set under LOO. L_2 is closest to the diagonal (ρ=0.87); L_1 and L_3 are noisier but rank-significant (ρ=0.56, 0.59).

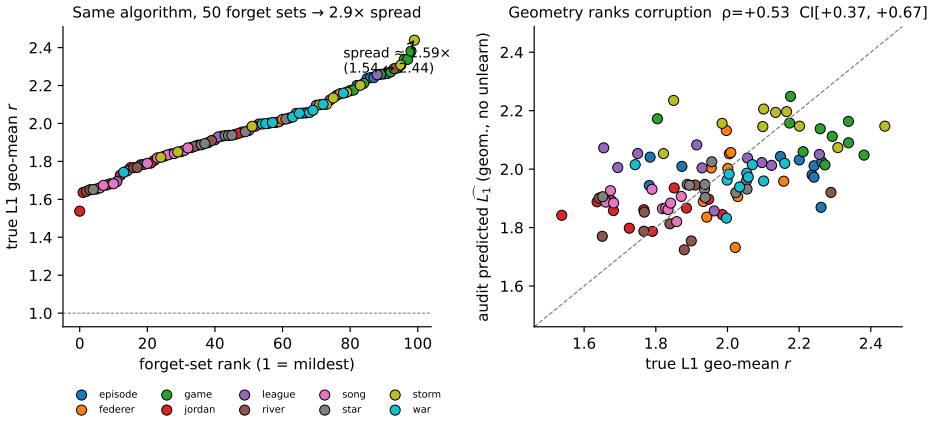

Figure 5: **Forget-set spread.** Distribution of per-forget-set L_1 across the 100 triplets, grouped by HDBSCAN cluster. The variation auditing must capture is between-cluster more than within-cluster.

Take-away. The audit predictor is unlearner-agnostic: the 12-dim forget-set descriptor lifts all three layers above the LOO-mean baseline for both GradAscent and NPO under independent training. Top-10% recall is in fact higher on NPO ($8/10$ on L_2, $5/10$ on L_3), likely because NPO's narrower dynamic range makes the ranking task easier. The same conclusion is reached separately per unlearner without label transfer between them.

5.6 Positioning: a Pre-Screen, Not a Replacement

We position the auditor as a *warning tool* for the forget-set selection step, not as a substitute for the actual unlearning evaluation.

Workflow. Given N candidate forget sets, audit all N in seconds, sort by predicted impact profile, and run real unlearning only on the top-k. At N=100 and k=10, the pre-screen reduces the unlearning budget by 10× while retaining the worst-case offenders identified by the auditor.

	Can	Cannot
Rank	flag "which $\mathcal{D}_f$ is worse" (ρ CIs > 0, L_2 0.87)	assign trustworthy absolute PPL values
Screen	red/yellow/green warnings on candidates	score an unlearning algorithm
Compute	seconds vs. several GPU-hours	replace the final unlearn run

Table 6: Audit positioning: cheap pre-screen for forget-set selection.

6 Discussion

6.1 What the geometric features actually capture

A natural question is why 12 scalar features computed in milliseconds carry enough signal to rank the locality damage with ρ=0.87. We hypothesise three mechanisms, each consistent with the per-cluster pattern in Fig. 3:

- **Concentration ⇒ stronger locality damage.** Forget sets with low spread and high

pairwise similarity (e.g. the "storm" cluster) sit on a tight manifold; the gradient signal from GradAscent is therefore sharply directional, and validation/test neighbours along the same direction are dragged with it. This predicts large L_2, which we observe.

- **Effective rank ⇒ spillover footprint**. Low-effective-rank $\mathcal{D}_f$ lives near a low-dim subspace; updating the model along that subspace bleeds across topic clusters that share the same dominant directions, producing $L_3 > 1$. The (negative) correlation between subspace coverage and L_3 ($\rho = -0.48$) supports this view.
- **Centroid norm ⇒ baseline PPL coupling**. Forget sets far from the embedding origin tend to have higher base PPL, which inflates the $r = \mathrm{PPL}_{\text{unlearned}}/\mathrm{PPL}_{\text{base}}$ ratio's denominator unevenly across layers; capturing this with `centroid_norm` corrects for the floor.

A causal study — e.g. deliberately constructing $\mathcal{D}_f$ with prescribed isotropy and effective rank — is left to future work; the present paper establishes the predictive correlation under naturally occurring HDBSCAN clusters.

6.2 When does the auditor break?

The audit is most reliable for L_2 and least reliable for L_3 in absolute terms (RF MAE 0.019 vs. baseline 0.022 — a 14% improvement, with $\rho = 0.59$ and 95% CI $[+0.44, +0.71]$). Two reasons: (i) L_3 has a very narrow dynamic range ($1.13\times$ between worst and mildest cluster), so absolute prediction is intrinsically hard; and (ii) cross-cluster spillover plausibly depends on *evaluation-side* geometry (where the neighbours of $\mathcal{D}_f$ are in the broader embedding space), which the forget-set-only descriptor cannot see. Top-1 identification is unstable in the same-cluster regime (multiple triplets within a cluster have very similar predicted scores), which is why we frame the auditor as a top-k screen rather than a precise ranker.

7 Conclusion

We formalised unlearning impact as a three-layer profile (L_1 forget, L_2 locality, L_3 spillover) and showed that, under fixed algorithm and hyperparameters, the profile depends systematically on the forget set: spillover does not vanish even at cross-topic distance, and damage magnitude varies measurably across candidates. A 12-dimensional Random Forest auditor predicts the profile from forget-set geometry alone, and the same conclusion holds under an independent NPO replication. This supports a screen-then-run workflow, cheap pre-screen of candidate forget sets, full unlearning only on the riskiest.

8 Related Work

8.1 LLM Unlearning

Modern LLM unlearning uses post-training updates to suppress targeted data while preserving the rest of the model. Representative methods include *Gradient Ascent* and gradient-difference variants (Jang et al., 2023), *NPO* (Negative Preference Optimisation) (Zhang et al., 2024), and representation misdirection (Li et al., 2024). Benchmarks such as TOFU (Maini et al., 2024), MUSE (Shi et al., 2024), and WMDP (Li et al., 2024) fix specific forget splits and report aggregate forget-quality and utility scores. The OpenUnlearning framework (Dorna et al., 2025) unifies many algorithms and metrics under one trainer. These works mainly ask which algorithm performs best for a given benchmark forget split; our work instead fixes the algorithm and studies how the choice of $\mathcal{D}_f$ reshapes the impact profile.

8.2 Evaluating Collateral Damage in Unlearning

Unlearning is increasingly recognised as collateral-damaging. Ko et al. (2025) introduce *knowledge-hole probing*, showing that standard utility benchmarks can miss adjacent and latent knowledge silently damaged by unlearning. Related model-editing and influence-function work also shows that gradient updates can have locality and non-locality effects (Meng et al., 2022; Mitchell et al., 2022; Koh and Liang, 2017; Grosse et al., 2023). Our three-layer view operationalises this concern as a dataset-level measurement: L_1 captures intended degradation, L_2 captures same-topic locality damage, and L_3 captures cross-topic spillover. Unlike post-hoc probes, our audit uses this measurement as the target for a pre-unlearning predictor.

8.3 Forget-Set Selection and Auditing

A small body of work studies which data is hardest to remove or causes the most damage when removed (Thudi et al., 2022; Kurmanji et al.,

2023). Influence-based selection is related (Koh and Liang, 2017), but it requires model gradients, scales poorly to 8B-parameter LLMs, and usually produces per-example scores rather than a forget-set-level impact profile. A separate data-first line predicts downstream model behaviour from dataset properties (Dang et al., 2024; Ilyas et al., 2022; Yao et al., 2020). We connect these threads by predicting the three-layer unlearning impact profile from forget-set geometry alone: the auditor uses only 50 forget-set embeddings and 12 scalar features, and serves as a cheap pre-screen before full unlearning evaluation.

9 Limitations

We acknowledge three "singles" that the present study does not control for:

1. **Single base model** (Llama-3.1-8B-Instruct). Geometry-to-corruption mapping may be model-specific; verifying on Mistral-7B and a Qwen variant is high-priority next work.
2. n=100. All three CIs exclude zero on both unlearners (Sec. 5.5), but L_3 remains the weakest layer; scaling to n=500–1000 is feasible in one extra GPU-day.
3. **Single metric**: PPL ratio. PPL captures token-level corruption but not task-level utility (QA accuracy). Parallel QA-based evaluation is in progress as a complementary axis.

GradDiff and RMU are the natural next unlearners to add alongside the GradAscent and NPO pair already covered.

References

Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. 2013. Density-based clustering based on hierarchical density estimates. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*.

Cuong Dang, Dung D. Le, and Thai Le. 2024. A curious case of searching for the correlation between training data and adversarial robustness of transformer textual models. *arXiv preprint arXiv:2402.11469*.

Vineeth Dorna, Anmol Mekala, Wenlong Zhao, Andrew McCallum, Zachary C. Lipton, J. Zico Kolter, and Pratyush Maini. 2025. OpenUnlearning: Accelerating LLM unlearning via unified benchmarking of methods and metrics. *arXiv preprint arXiv:2506.12618*.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, and 1 others. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Roger Grosse, Juhan Bae, Cem Anil, and 1 others. 2023. Studying large language model generalization with influence functions. *arXiv preprint arXiv:2308.03296*.

Andrew Ilyas, Sung Min Park, Logan Engstrom, Guillaume Leclerc, and Aleksander Madry. 2022. Datamodels: Predicting predictions from training data. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*.

Joel Jang, Dongkeun Yoon, Sohee Yang, Sungmin Cha, Moontae Lee, Lajanugen Logeswaran, and Minjoon Seo. 2023. Knowledge unlearning for mitigating privacy risks in language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*.

Myeongseob Ko, Hoang Anh Just, Charles Fleming, Ming Jin, and Ruoxi Jia. 2025. Probing knowledge holes in unlearned LLMs. *arXiv preprint arXiv:2511.00030*.

Pang Wei Koh and Percy Liang. 2017. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*.

Meghdad Kurmanji, Peter Triantafillou, Jamie Hayes, and Eleni Triantafillou. 2023. Towards unbounded machine unlearning. *Advances in Neural Information Processing Systems (NeurIPS)*.

Nathaniel Li, Alexander Pan, Anjali Gopal, and 1 others. 2024. The WMDP benchmark: Measuring and reducing malicious use with unlearning. *arXiv preprint arXiv:2403.03218*.

Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. 2024. TOFU: A task of fictitious unlearning for LLMs. *arXiv preprint arXiv:2401.06121*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*.

Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D. Manning, and Chelsea Finn. 2022. Memory-based model editing at scale. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*.

Weijia Shi, Jaechan Lee, Yangsibo Huang, Sadhika Malladi, Jieyu Zhao, Ari Holtzman, Daogao Liu, Luke Zettlemoyer, Noah A. Smith, and Chiyuan

Zhang. 2024. MUSE: Machine unlearning six-way evaluation for language models. *arXiv preprint arXiv:2407.06460*.

Anvith Thudi, Gabriel Deza, Varun Chandrasekaran, and Nicolas Papernot. 2022. Unrolling SGD: Understanding factors influencing machine unlearning. In *IEEE European Symposium on Security and Privacy (EuroS&P)*.

Wenhui Wang, Hangbo Bao, Shaohan Huang, Li Dong, and Furu Wei. 2021. MiniLMv2: Multi-head self-attention relation distillation for compressing pretrained transformers. In *Findings of ACL*.

Zhewei Yao, Amir Gholami, Kurt Keutzer, and Michael W. Mahoney. 2020. PyHessian: Neural networks through the lens of the hessian. In *IEEE International Conference on Big Data*.

Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. 2024. Negative preference optimization: From catastrophic collapse to effective unlearning. *arXiv preprint arXiv:2404.05868*.

A Reproducibility Details

Data. We filter WikiText-103 to 16,038 passages with at least 32 tokens. MiniLM-L6-v2 embeddings are used with all 384 dimensions kept.

Component	Setting
HDBSCAN	min cluster 200; min samples 5
Distance	Euclidean; EOM selection
Cluster sizes	202/346/3,276 (min/median/max)
Triplets	100 triplets, seed 42
Split size	50 texts each; disjoint within triplet
Eligibility	cluster size at least 150

Unlearning. The OpenUnlearning script is:

```
wikitext_unlearn_tofu_aligned.sh
```

The run configuration is:

Hyperparameter	Value
Epochs	5
Batch / accumulation	8/8 (effective batch 64)
Learning rate	1×10^{-5}, linear decay to zero
Warmup / weight decay	1/0.01
Optimizer / precision	paged AdamW 32-bit / BF16

With $|\mathcal{D}_f|$=50 and effective batch 64, exactly 5 optimiser steps fire (one per epoch); train loss ranges from -0.84 to -0.62.

Evaluation. PPL is computed in BF16 with stride 512 and context 1024; each text is scored independently. Base-model PPL is cached once per evaluation triplet and split, then reused across the 100 unlearned checkpoints. The long-format output is `ppl_long.parquet`, with one row per (m, e, s, i) tuple.

Audit. The feature script `features/forget_set.py` computes the 12 scalars from a 50×384 embedding matrix in <10 ms. The auditor uses scikit-learn's Random Forest regressor:

Setting	Value
Trees	200
Minimum leaf size	2
Maximum depth	None
Seed	0
Evaluation	LOO over 100 forget sets
Bootstrap	10,000 resamples, seed 0

Hardware. All runs use a single H100 (80 GB) node.

Stage	Wall-clock
Data preparation	$\approx$ 20 min
100 unlearning runs	1 h 54 min
Cross-PPL	6 h 11 min

B Additional Sanity Tables

Per-cluster monotonicity. Across all 10 clusters, $L_1(c)>L_2(c)>L_3(c)$ holds without exception. The smallest within-cluster gap L_1-L_2 is 0.39 (mildest cluster), and the largest is 0.71 (storm).

Per-cluster three-layer profile. Table 7 reports cluster-level mean and within-cluster spread (max/min over the 10 triplets in that cluster) for each of the three layers. **Within-cluster spread is non-trivial in every cluster on every layer** (min 1.02, max 1.39), so the global spread 1.59/1.50/1.13 in Table 2 is not an artefact of pooling across clusters: a single cluster's 10 triplets already exhibit measurable forget-set-induced variation. Cluster means also vary (L_1: 1.78–2.22; L_2: 1.20–1.59; L_3: 1.16–1.23), but the within-cluster signal alone is sufficient to support C3.

The full per-triplet (L_1, L_2, L_3) table is released as supplementary CSV (`part1_corruption_profile.csv`).

	L_1		L_2		L_3	
Cluster	mean	spread	mean	spread	mean	spread
1	2.22	1.32	1.47	1.08	1.20	1.02
2	2.00	1.12	1.26	1.07	1.19	1.05
3	1.78	1.29	1.31	1.10	1.21	1.07
4	2.10	1.27	1.37	1.09	1.20	1.04
5	1.94	1.36	1.38	1.17	1.21	1.09
6	1.79	1.13	1.20	1.08	1.16	1.05
7	2.02	1.24	1.21	1.06	1.19	1.05
8	2.11	1.34	1.59	1.19	1.23	1.05
9	1.83	1.39	1.24	1.08	1.18	1.08
10	1.93	1.24	1.24	1.05	1.16	1.06
All 100	1.97	**1.59**	1.33	**1.50**	1.19	**1.13**

Table 7: Per-cluster three-layer profile (n=10 triplets per cluster). Spread is within-cluster max/min. Last row pools all 100 forget sets and reports the global spread reported in the main text. The fact that spread > 1 holds inside every individual cluster confirms that forget-set-induced variation is not driven by between-cluster differences alone.